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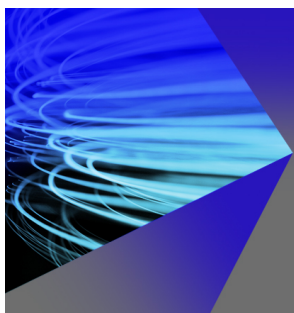
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Spherical wave expansion for any spin

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The spherical wave expansion is derived for fields of particles of arbitrary spin and mass, and for arbitrary helicity, massless particles (such as the electromagnetic vector potential corresponding to photons). The starting point is Weinberg's characterization of relativistic, higher spin fields that transform according to general irreducible representations of the homogeneous Lorentz group. The expansion is a relativistic generalization of the familiar tensor and spinor spherical harmonics. It is useful for central force problems that arise, for example, in scattering processes off spherically symmetric solitonic backgrounds such as monopoles or skyrmions.

I. INTRODUCTION

The importance of partial wave expansions in physics cannot be overstated. For quantum mechanics, in particular, it provides a framework for intuition about processes and also a simplifying calculational scheme. Spherical partial waves are the basis in which the consequences of angular momentum conservation about a spherically symmetric potential are manifest. From a computational standpoint, the original hyperbolic, partial differential Schrödinger equation is separated in the spherical basis. The separated wave function has a specific, known angular dependence and a radial dependence that is determined by a much simpler, radial ordinary differential equation.¹

This expansion is typically applied to nonrelativistic systems where, in terms of symmetries, the fields transform as representations of the Galileo group. Spin is then particularly easy to work with because it comprises a separate, independent invariance. In contrast, for relativistic fields, this simplification no longer holds (boosts and rotations do not commute). The resulting spherical waves are then more complicated. For example, in the case of Dirac spin-1/2 electrons, the partial waves for the electron field are not diagonal in orbital angular momentum. A wave with definite frequency can also have a definite total angular momentum, J , but it will involve a sum of spherical harmonics, $Y_{J+1/2,m}(\hat{x})$ and $Y_{J-1/2,m}(\hat{x})$. This is typically interpreted as meaning that relativistic electrons do not have well-defined orbital angular momentum.² A more precise interpretation, which is made clear in the analysis presented in this article, is that higher spin particle states may be resolved into states with definite orbital angular momentum, but the field modes used to describe these states are not sharp in orbital angular momentum spherical harmonics.

This distinction is academic in the case of relativistic, spin-1/2 electrons subject to electromagnetic interactions that are parity invariant. For then, only two orbital angular momenta, $J \pm 1/2$, are possible in a scattering channel that has total angular momentum J . The parity invariant interactions will not induce a change of one unit in orbital angular momentum. The theoretical S matrix will then be diagonal in orbital angular momentum, and the correspondence to experimentally determined partial wave amplitudes is straightforward.³ In general, however, the partial wave amplitudes for particles with spin will not be diagonal in orbital angular momentum. Then the following problem arises: How are the partial wave amplitudes, which themselves are not diagonal in orbital angular momentum, embedded in the nondiagonal partial wave field modes? To be more specific, consider an example where the scattering is described by a model with coupled, linear, Schrödinger-like equations for the fields of the higher spin particles. And suppose that the equations are spherically symmetric, thus describing a generalized central force problem. The distorted wave Born approximation (DWBA) gives rise to many such problems, where the distorted waves satisfy Schrödinger-like equations. Quanta-soliton scattering is another common example, where

the fields satisfy linearized equations about a spherically symmetric solitary wave background.^{4–6} The problem then is how to exploit the spherical symmetry to separate these equations into simpler, coupled radial equations.

We will see that two assumptions about the theory will be sufficient to unambiguously answer this question. First, it is assumed that the theory is Poincaré invariant, and that the fields transform as general irreducible representations of the homogeneous Lorentz group. Second, the spherically symmetric potential is assumed to vanish, or falloff, as $r \rightarrow \infty$ sufficiently fast so that the scattering modes are asymptotic to free modes. I will not derive bounds on the falloff behavior, and instead refer the reader to Refs. 6 and 7 for examples of sufficiently rapid falloff of the potential.

Of course, the spherical wave expansion for relativistic fields is useful even for potentials with bound states, such as the Coulomb potential. In principle, the expansions presented in this article may be easily generalized to accommodate the presence of bound waves. Basically, the radial modes decay exponentially rather than oscillating out to infinity. Although this modification is conceptually trivial, it necessarily introduces additional notation to an already complicated mess. Therefore I will restrict my attention to potentials with no bound states.

In the next section, I briefly review Weinberg's plane wave expansion for free relativistic fields of arbitrary spin.⁸ This formalism accommodates fields for massive and massless higher spin particles. Then in Sec. III, I derive the spherical wave expansion for the free and scattered fields of higher spin and mass $m > 0$. This involves a separation of variables that is appropriate for scattering off spherically symmetric potentials. The answer to the above posed question is thus contained in this section. In Sec. IV, I derive the corresponding results for the mass $m = 0$ case, both for free and scattered fields. Then in Sec. V, I present the expansion for the case of tensor potentials which are the arbitrary spin or helicity generalization of the electromagnetic vector potential (and are not to be confused with the previously referred to potentials that appear in the Schrödinger-like equations that the fields satisfy). Finally, Sec. VI concludes the article by recovering the well-known spherical wave expansions of the electromagnetic vector potential and of the spin-1/2 Dirac electron from the general expansions given in Secs. V and III, respectively.

II. FREE FIELD PLANE WAVE EXPANSION

This section is a brief review of that subset of results in Weinberg's plane wave expansion of relativistic, higher spin free fields⁸ that I will use to derive the corresponding spherical wave expansion (SWE). This review will closely follow some unpublished lecture notes of Weinberg.⁹ The basic requirement of the fields for particles of arbitrary spin is that they transform as irreducible representations of the homogeneous Lorentz group (HLG). I will first run through Wigner's little group construction of particle states of arbitrary spin.¹⁰ Then the requirement that the corresponding field transforms covariantly will impose a covariance condition on the wave functions. Using some representation theory of the HLG, we may solve this condition and determine the wave functions. Both the mass, $m > 0$, case and the zero mass, $m = 0$, case will be worked out. For more details and motivation, see Refs. 8 and 10.

First consider the $m > 0$ case. In Wigner's little group construction, a particle of spin s with $\hat{z}$ component equal to σ , and moving with momentum $\mathbf{p}$ is defined by

$$|\mathbf{p}, \sigma\rangle = \sqrt{\frac{m}{E_p}} U(L(\mathbf{p}/m)) |0, \sigma\rangle. \quad (2.1)$$

Here $|0, \sigma\rangle$ is the fiducial, $m > 0$ state with the particle at rest and spin $\hat{z}$ component equal to σ . $U(L(\mathbf{p}/m))$ is a unitary operator acting on states and it implements the Lorentz transformation $L(\mathbf{p}/m)$ defined below in Eqs. (2.3)–(2.5). The energy of the moving particle is $E_p = \sqrt{\mathbf{p}^2 + m^2}$.

The definition in Eq. (2.1) may be understood by considering the $m>0$ little group, which is defined to be the subset of the HLG that leaves the fiducial particle momentum unchanged. For $m>0$, the little group is the set of rotations. The states $|\mathbf{0}, \sigma\rangle$ comprise an irreducible, spin s representation of rotations.¹¹ Hence for a rotation R

$$U(R)|\mathbf{0}, \sigma\rangle = \sum_{\sigma'} D_{\sigma'\sigma}^{(s)}(R)|\mathbf{0}, \sigma'\rangle. \quad (2.2)$$

To get a spin s particle with nonzero momentum, Wigner's convention is to boost up using the HLG transformation $L(\mathbf{p}/m)$ defined by

$$L(\mathbf{p}/m) = R(\hat{\mathbf{p}})B(|\mathbf{p}|/m)R^{-1}(\hat{\mathbf{p}}), \quad (2.3)$$

where $B(|\mathbf{p}|/m)$ is a boost along the $\hat{\mathbf{z}}$ axis from rest to rapidity θ where $|\mathbf{p}|/m = \sinh \theta$. So

$$B(|\mathbf{p}|/m) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cosh \theta & \sinh \theta \\ 0 & 0 & \sinh \theta & \cosh \theta \end{pmatrix}. \quad (2.4)$$

Also, $R(\hat{\mathbf{p}})$ is a rotation that takes $\hat{\mathbf{z}}$ into $\hat{\mathbf{p}}$

$$R(\hat{\mathbf{p}})^i_j \hat{\mathbf{z}}^j = \hat{\mathbf{p}}^i. \quad (2.5)$$

For definiteness, I will take $R(\hat{\mathbf{p}}) = \mathcal{R}(\phi, \theta, 0)$ where $\mathcal{R}(\alpha, \beta, \gamma)$ denotes a rotation with Euler angles α, β, γ in the conventions of Beidenharn and Louck,¹¹ and ϕ and θ are the spherical angles of the unit vector, $\hat{\mathbf{p}}$.

From Eq. (2.1), we conclude that the states transform under the HLG transformation Λ as

$$U(\Lambda)|\mathbf{p}, \sigma\rangle = \sqrt{\frac{E_{\Lambda p}}{E_p}} \sum_{\sigma'} D_{\sigma'\sigma}^{(s)}(W(\Lambda, \mathbf{p}/m))|\Lambda\mathbf{p}, \sigma'\rangle, \quad (2.6)$$

where $W(\Lambda, \mathbf{p}/m)$ is the Wigner rotation given by

$$W(\Lambda, \mathbf{p}/m) = L^{-1}(\Lambda\mathbf{p}/m)\Lambda L(\mathbf{p}/m). \quad (2.7)$$

Note that $W(\Lambda, \mathbf{p}/m)$ is in the $m>0$ little group—that is, $W(\Lambda, \mathbf{p}/m)$ is a rotation. Further, when Λ is a simple rotation, say, $\Lambda = R$, then the corresponding Wigner rotation is equal to R

$$W(R, \mathbf{p}/m) = R. \quad (2.8)$$

Thus, the states $|\mathbf{p}, \sigma\rangle$ transform under rotations as one would naively expect from nonrelativistic quantum mechanics. In particular, the states may be projected onto states of definite orbital angular momentum, l , together with total angular momentum, J , and $\hat{\mathbf{z}}$ -component J_z , according to

$$|EJJ_zls\rangle = \sum_{m\sigma} \langle lsm\sigma | JJ_z \rangle \sqrt{pE} \int d^2\hat{\mathbf{p}} Y_{lm}(\hat{\mathbf{p}})|\mathbf{p}, \sigma\rangle. \quad (2.9)$$

To repeat, the orbital angular momentum is a good quantum number for relativistic states in the continuum and it combines with spin s as in Eq. (2.9) as a direct product to form total spin J states.

Finally, denoting the unitary operator for a space–time translation by a^μ as $U(a)$ we have

$$U(a)|\mathbf{p}, \sigma\rangle = e^{-ip \cdot a}|\mathbf{p}, \sigma\rangle. \quad (2.10)$$

Next, consider the case of a massless particle, $m=0$, with helicity λ . Wigner defines the state for this particle with momentum $\mathbf{p}$ and helicity λ to be

$$|\mathbf{p}, \lambda\rangle = \sqrt{\frac{\kappa}{|\mathbf{p}|}} U(\mathcal{L}(\mathbf{p}/\kappa))|\lambda\rangle. \quad (2.11)$$

Here $|\lambda\rangle$ is the state where the particle has a fiducial four-momentum $\kappa^\mu = (0, 0, \kappa, \kappa)$. It is moving in the $\hat{z}$ direction with some fixed but arbitrary frequency κ . The unitary operator $U(\mathcal{L}(\mathbf{p}/\kappa))$, where $\mathcal{L}(\mathbf{p}/\kappa)$ is a Lorentz transformation defined below in Eqs. (2.22)–(2.24), transforms the fiducial state up to arbitrary momentum $\mathbf{p}$. The corresponding little group consists of rotations about the $\hat{z}$ axis, and also simultaneous $\hat{x}$ boosts and $\hat{y}$ rotations or $\hat{y}$ boosts and $\hat{x}$ rotations.

To be more precise, consider infinitesimal HLG transformations

$$\Lambda_\nu^\mu = \delta_\nu^\mu + \omega_\nu^\mu. \quad (2.12)$$

The requirement

$$\Lambda_\nu^\mu \kappa^\nu = \kappa^\mu \quad (2.13)$$

then has a three parameter solution specified by

$$\omega^{12} = -\omega^{21} = \theta, \quad \omega^{10} = -\omega^{01} = \omega^{13} = -\omega^{31} = a, \quad \omega^{20} = -\omega^{02} = \omega^{23} = -\omega^{32} = b. \quad (2.14)$$

It will be useful to have the Lie algebra that generates these transformations. Consider then

$$U(1 + \omega) = 1 + \frac{i}{2} \omega_{\mu\nu} J^{\mu\nu} \quad (2.15)$$

and define rotation generators, J_i , and boost generators, K_i , by

$$J_i \equiv \frac{1}{2} \epsilon_{ijk} J_{jk}, \quad K_i \equiv J_{i0} = -J_{0i}. \quad (2.16)$$

They satisfy the commutation relations

$$[J_i, J_j] = i \epsilon_{ijk} J_k, \quad [J_i, K_j] = i \epsilon_{ijk} K_k, \quad [K_i, K_j] = -i \epsilon_{ijk} J_k. \quad (2.17)$$

From Eq. (2.14) it follows that the generators of the little group elements are J_3 , L_1 , and L_2 where

$$L_1 \equiv K_1 - J_2, \quad L_2 \equiv K_2 + J_1. \quad (2.18)$$

Their commutators are

$$[J_3, L_1] = iL_2, \quad [J_3, L_2] = -iL_1, \quad [L_1, L_2] = 0. \quad (2.19)$$

The generators L_1 and L_2 thus form an invariant Abelian subalgebra. It follows that their only finite dimensional representation in the little group is trivial,⁸ and thus we will require that the states $|\lambda\rangle$ satisfy

$$L_1|\lambda\rangle = L_2|\lambda\rangle = 0, \quad J_3|\lambda\rangle = \lambda|\lambda\rangle. \quad (2.20)$$

Exponentiating, we see that the physically desirable, finite dimensional irreducible representations of the $m=0$ little group element $\mathcal{R}$ are all one dimensional

$$U(\mathcal{R}) = e^{i\lambda\Theta[\mathcal{R}]}|\lambda\rangle. \quad (2.21)$$

This is the $m=0$ analog of Eq. (2.2).

To get a helicity λ particle with nonzero momentum we boost $|\lambda\rangle$ as in Eq. (2.11), where

$$\mathcal{L}(\mathbf{p}/\kappa) = R(\hat{p})\mathcal{B}(|\mathbf{p}|/\kappa). \quad (2.22)$$

$R(\hat{p})$ rotates $\hat{z}$ into $\hat{p}$ and is defined as in the massive case. $\mathcal{B}(|\mathbf{p}|/\kappa)$ is a boost in the $\hat{z}$ direction from frequency κ to $|\mathbf{p}|$. Thus

$$\mathcal{B}(|\mathbf{p}|/\kappa) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cosh \phi & \sinh \phi \\ 0 & 0 & \sinh \phi & \cosh \phi \end{pmatrix}, \quad (2.23)$$

where

$$\phi = \ln(|\mathbf{p}|/\kappa). \quad (2.24)$$

From the definition (2.11), it follows that the $m=0$, helicity λ states transform as

$$U(\Lambda)|\mathbf{p}, \lambda\rangle = \sqrt{\frac{|\Lambda\mathbf{p}|}{|\mathbf{p}|}} e^{i\lambda\Theta[\mathcal{W}(\Lambda, \mathbf{p}/\kappa)]} |\Lambda\mathbf{p}, \lambda\rangle, \quad (2.25)$$

where $\Theta[\mathcal{W}(\Lambda, \mathbf{p}/\kappa)]$ is the $\hat{z}$ rotation angle corresponding to the Wigner rotation

$$\mathcal{W}(\Lambda, \mathbf{p}/\kappa) = \mathcal{L}^{-1}(\Lambda\mathbf{p})\Lambda\mathcal{L}(\mathbf{p}). \quad (2.26)$$

Also note that the states $|\mathbf{p}, \lambda\rangle$ transform under translations as in Eq. (2.10).

Massless free states, like massive states, may be chosen to have definite angular momentum. In contrast to the $m>0$ definition in Eq. (2.9), the normalized helicity λ state with frequency p , total angular momentum J , and $\hat{z}$ component J_z is defined by

$$|pJJ_z\lambda\rangle \equiv p \sqrt{\frac{2J+1}{4\pi}} \int d^2\hat{p} D_{J_z\lambda}^{(J)*}(R(\hat{p}))|\mathbf{p}\lambda\rangle. \quad (2.27)$$

To see that this state rotates properly, first note that the rotational properties of the helicity states in Eq. (2.25) together with the definition of the rotation matrices in Ref. 11 imply that

$$D_{J_z\lambda}^{(J)*}(R(\hat{p}))|\mathbf{p}\lambda\rangle = D_{J_z\lambda}^{(J)*}(\mathcal{R}(\phi, \theta, 0))\mathcal{R}(\phi, \theta, 0)|p\hat{z}, \lambda\rangle = D_{J_z\lambda}^{(J)*}(\mathcal{R}(\phi, \theta, \psi))\mathcal{R}(\phi, \theta, \psi)|p\hat{z}, \lambda\rangle \quad (2.28)$$

for any angle ψ . Thus

$$|pJJ_z\lambda\rangle = p \sqrt{\frac{2J+1}{4\pi}} \frac{1}{2\pi} \int d\phi d\theta d\psi \sin\theta D_{J_z\lambda}^{(J)*}(\mathcal{R}(\phi, \theta, \psi))\mathcal{R}(\phi, \theta, \psi)|p\hat{z}, \lambda\rangle. \quad (2.29)$$

But then, rotational invariance of the $SU(2)$ measure implies that

$$\begin{aligned}
\mathcal{R}(\alpha, \beta, \gamma) |pJJ_z\lambda\rangle &= p \sqrt{\frac{2J+1}{4\pi}} \frac{1}{2\pi} \int d\phi d\theta d\psi \sin\theta \\
&\quad \times D_{J\lambda}^{(J)*}(\mathcal{R}(\phi, \theta, \psi)) \mathcal{R}(\alpha, \beta, \gamma) \mathcal{R}(\phi, \theta, \psi) |p\tilde{x}\lambda\rangle \\
&= p \sqrt{\frac{2J+1}{4\pi}} \frac{1}{2\pi} \int d\phi' d\theta' d\psi' \sin\theta' \\
&\quad \times D_{J\lambda}^{(J)*} \mathcal{R}^{-1}(\alpha, \beta, \gamma) (\mathcal{R}(\phi', \theta', \psi')) \mathcal{R}(\phi', \theta', \psi') |p\tilde{z}\lambda\rangle \\
&= D_{J_z' J_z}^{(J)}(\mathcal{R}(\alpha, \beta, \gamma)) |pJJ_z'\lambda\rangle
\end{aligned} \tag{2.30}$$

as is desired.

Now that the free particle states and their transformation properties are specified, we may evaluate the consequences of Lorentz covariance on the corresponding free fields. For our purposes, it will be sufficient to focus on the positive frequency part of the fields. I will comment, when relevant, on the modifications that hold for the negative frequency components. So let $\phi_n(x)$ be a field corresponding to a mass $m > 0$, spin s particle. The basic covariance requirement is that the field transform under a HLG transformation Λ and translation by a^μ as

$$U(\Lambda, a) \phi_n(x) U^{-1}(\Lambda, a) = \sum_m D_{nm}(\Lambda^{-1}) \phi_m(\Lambda x + a). \tag{2.31}$$

First consider translations. From Eq. (2.10) it follows that the annihilation operator, $a(\mathbf{p}, \sigma)$, transforms as

$$U(a) a(\mathbf{p}, \sigma) U^{-1}(a) = e^{ip \cdot a} a(\mathbf{p}, \sigma). \tag{2.32}$$

Thus, translational covariance of the field $\phi_n(x)$ in Eq. (2.31) may be realized by Fourier expanding

$$\phi_n(x) = \sum_\sigma \int \frac{d^3p}{(2\pi)^{3/2} \sqrt{2E_p}} u_n(\mathbf{p}, \sigma) e^{ip \cdot x} a(\mathbf{p}, \sigma). \tag{2.33}$$

The factor $(2E)^{-1/2}$ is included to simplify the covariance constraint on the “wave function” $u_n(\mathbf{p}, \sigma)$.

Next consider the HLG transformations. From Eqs. (2.6) and (2.33), we conclude

$$U(\Lambda) a(\mathbf{p}, \sigma) U^{-1}(\Lambda) = \sqrt{\frac{E_{\Lambda p}}{E_p}} \sum_{\sigma'} D_{\sigma'\sigma}^{(s)}(W(\Lambda, \mathbf{p}/m)) a(\Lambda \mathbf{p}, \sigma'). \tag{2.34}$$

But then, for Eqs. (2.34) and (2.31) to be consistent we must have

$$\sum_\sigma D_{\sigma'\sigma}^{(s)*}(W(\Lambda, \mathbf{q}/m)) u_n(\mathbf{q}, \sigma) = \sum_m D_{nm}(\Lambda^{-1}) u_m(\Lambda \mathbf{q}, \sigma'). \tag{2.35}$$

The explicit solution of this fundamental condition requires a little bit of the theory of HLG representations. The same algebra used to prove Eqs. (2.15)–(2.17) for the generators $J^{\mu\nu}$ holds for their finite dimensional matrix representations. If we define

$$\mathcal{A}_i = \frac{1}{2}(J_i + iK_i), \quad \mathcal{B}_i = \frac{1}{2}(J_i - iK_i) \tag{2.36}$$

then Eq. (2.17) implies that

$$[\mathcal{A}_i, \mathcal{A}_j] = i\epsilon_{ijk}\mathcal{A}_k, \quad [\mathcal{B}_i, \mathcal{B}_j] = i\epsilon_{ijk}\mathcal{B}_k, \quad [\mathcal{A}_i, \mathcal{B}_j] = 0. \quad (2.37)$$

Thus $\mathcal{A}$ and $\mathcal{B}$ generate two commuting SU(2) groups. We may label the finite dimensional, irreducible representations by (A, B) , where the generators are

$$(\mathcal{A})_{a'b', ab} = \delta_{b'b}\mathbf{J}_{a'a}^{(A)}, \quad (\mathcal{B})_{a'b', ab} = \delta_{a'a}\mathbf{J}_{b'b}^{(B)}. \quad (2.38)$$

Here, $\mathbf{J}_{a'a}^{(A)}$ is the usual triplet of spin A matrices

$$(\mathbf{J}_3^{(A)})_{a'a} = a\delta_{a'a}, \quad (\mathbf{J}_1^{(A)} \pm i\mathbf{J}_2^{(A)})_{a'a} = \delta_{a', a \pm 1} \sqrt{(A \mp a)(A \pm a + 1)}. \quad (2.39)$$

Take the field ϕ_n to be in the (A, B) representation. Since

$$\mathbf{J} = \mathbf{A} + \mathbf{B} \quad (2.40)$$

(A, B) must be such that the triangle inequality with spin s holds

$$s = A + B, A + B - 1, \dots, |A - B|. \quad (2.41)$$

Now, as with the constructions of the states, condition (2.35) is best solved by first focusing on the little group transformations (spatial rotations for $m > 0$), and then boosting up to arbitrary momentum. For $\mathbf{q} = \mathbf{0}$ and $\Lambda = R$, Eq. (2.35) implies

$$\sum_{\sigma} D_{\sigma\sigma'}^{(s)}(R) u_{ab}(\mathbf{0}, \sigma) = \sum_{a'b'} D_{aa'}^{(A)}(R) D_{bb'}^{(B)}(R) u_{a'b'}(\mathbf{0}, \sigma'). \quad (2.42)$$

This equation determines the embedding of the little group in the field representation. It couples $\mathbf{A}$ and $\mathbf{B}$ to form spin s . From standard angular momentum representation theory,¹¹ we find that Eq. (2.42) is satisfied by taking

$$u_{ab}(\mathbf{0}, \sigma) = \langle ABab | s\sigma \rangle. \quad (2.43)$$

Now to get the wave functions for nonzero momentum $\mathbf{p}$, use Eq. (2.35) with $\mathbf{q} = \mathbf{0}$ and $\Lambda = L(\mathbf{p}/m)$. A simple calculation gives

$$\begin{aligned} u_{ab}(\mathbf{p}, \sigma) &= \sum_{a'b'} (e^{-\theta\hat{\mathbf{p}} \cdot \mathbf{J}})_{aa'} (e^{\theta\hat{\mathbf{p}} \cdot \mathbf{J}})_{bb'} \langle ABa'b' | s\sigma \rangle \\ &= \sum_{a'b'\sigma'} e^{\theta(b' - a')} \langle ABa'b' | s\sigma' \rangle D_{\sigma\sigma'}^{(s)*}(R) D_{aa'}^{(A)}(R) D_{bb'}^{(B)}(R). \end{aligned} \quad (2.44)$$

In Eq. (2.44), $R = R(\hat{\mathbf{p}})$ as defined in Eq. (2.5), and $\sinh \theta = |\mathbf{p}|/m$. In the next section, we will see that the nontrivial dependence of the wave function $u_{ab}(\mathbf{p}, \sigma)$ on the momentum $\mathbf{p}$ will lead to a nondiagonal orbital angular momentum dependence in the spherical waves.

Next, consider the $m=0$, helicity λ case. As in the massive case, we construct the wave functions $u_{ab}(\mathbf{p}, \lambda)$ by solving the massless analog of Eq. (2.35)

$$e^{-i\lambda\Theta[\gamma^{-1}(\Lambda\mathbf{q}/\kappa)\Lambda\gamma(\mathbf{q}/\kappa)]} u_l(\mathbf{q}, \lambda) = \sum_m D_{lm}(\Lambda^{-1}) u_m(\Lambda\mathbf{q}, \lambda). \quad (2.45)$$

This is first solved for the little group transformations and then by boosting up to get the wave function for general momentum. So first let $q^\mu = \kappa^\mu$ and take $\Lambda = \mathcal{R} \in$ the little group. Then Eq. (2.45) implies

$$\sum_m D_{nm}(\mathcal{R}) u_m(\lambda) = e^{i\lambda\Theta(\mathcal{R})} u_n(\lambda). \quad (2.46)$$

For the fiducial momentum $\kappa^\mu = (0, 0, \kappa, \kappa)$, the little group generators are given in Eq. (2.19). Take $\mathcal{R}$ near the identity and expand it in terms of these generators. Using Eq. (2.36) and keeping terms linear in the generators, we find that Eq. (2.46) implies

$$\begin{aligned} (J_1^{(A)} - iJ_2^{(A)})_{aa'} u_{a'b}(\lambda) &= 0, \quad (J_1^{(B)} + iJ_2^{(B)})_{bb'} u_{ab'}(\lambda) = 0, \\ [(J_3^{(A)})_{aa'} \delta_{bb'} + (J_3^{(B)})_{bb'} \delta_{aa'}] u_{a'b'}(\lambda) &= \lambda u_{ab}(\lambda). \end{aligned} \quad (2.47)$$

These are solved by choosing (A, B) such that

$$\lambda = B - A \quad (2.48)$$

and by taking

$$u_{ab}(\lambda) = \delta_{a, -A} \delta_{b, B}. \quad (2.49)$$

Now to get the wave functions for general momentum $\mathbf{p}$, use Eq. (2.45) with $q^\mu = \kappa^\mu$ and $\Lambda = \mathcal{L}(\mathbf{p}/\kappa)$, where $\mathcal{L}(\mathbf{p}/\kappa)$ is defined in Eqs. (2.22)–(2.24). The solution is

$$u_{ab}(\mathbf{p}, \lambda) = D_{a, -A}^{(A)}(R) D_{b, B}^{(B)}(R) e^{(A+B)\phi} = D_{a, -A}^{(A)}(R) D_{b, B}^{(B)}(R) (|\mathbf{p}|/\kappa)^{A+B}. \quad (2.50)$$

Here, $R = R(\hat{\mathbf{p}})$ and $\phi = \ln|\mathbf{p}|/\kappa$.

Equations (2.44) and (2.50) explicitly determine the wave functions in terms of the momentum and the spin quantum numbers for the $m > 0$ and $m = 0$ cases, respectively. The covariance of the fields with these wave functions is thus ensured. The construction reviewed in this section is presented in several articles by Weinberg.⁸ The goal there was to develop a relativistic theory of higher spin fields. For example, a general Lorentz invariant Hamiltonian density may be simply built up out of polynomials of the above fields that are simultaneously $\mathcal{A}$ - and $\mathcal{B}$ -invariant. In this context, Weinberg goes on to give simple proofs of the necessity of antiparticles, crossing symmetry, spin statistics, and CPT invariance. These results are contained in the final result for the expansion of the full field, including the negative frequency components

$$\phi_{ab}(x) = \sum_{\sigma} \int \frac{d^3p}{(2\pi)^{3/2} \sqrt{2E_p}} u_{ab}(\mathbf{p}, \sigma) [e^{ip \cdot x} a(\mathbf{p}, \sigma) + (-1)^{B-A-\sigma} e^{-ip \cdot x} a_c^\dagger(\mathbf{p}, -\sigma)]. \quad (2.51)$$

Here, $a_c^\dagger(\mathbf{p}, \sigma)$ creates the antiparticle of the particle that is destroyed by $a(\mathbf{p}, \sigma)$. When $m = 0$, Eq. (2.51) holds, but with σ replaced by λ and Eq. (2.44) replaced by Eqs. (2.49) and (2.50). In the next two sections, we will find that another virtue of this approach to field theory is that it allows a direct construction of the SWE of higher spin fields.

III. SPHERICAL WAVE EXPANSION FOR $m > 0$

With the plane wave expansions given in the preceding section, it is in principle easy to transform to a spherical wave expansion (SWE). We just need to factorize the $\hat{p}$ dependences and integrate over $\hat{p}$. For spin $s > 0$ fields, the $\hat{p}$ dependence in the wave functions makes them not be

diagonal in the orbital angular momentum, spherical harmonic basis, $Y_{lm}(\hat{x})$. Nevertheless, the scalar, spin $s=0$ case will be worked out first. The notation is simpler there, thus allowing the lines of argument to be clearer. These same arguments will then admit an easy generalization to arbitrary spin. I will first work out the expansion for free fields, and then infer the expansion for the case of fields interacting with a spherically symmetric background.

For the $(A,B)=(0,0)$ scalar field, ϕ , Eqs. (2.44) and (2.33) imply that

$$\phi(x) = \int \frac{d^3p}{(2\pi)^{3/2}\sqrt{2E_p}} e^{ip \cdot x} a(\mathbf{p}, \sigma). \quad (3.1)$$

Now using the SWE for plane waves¹²

$$\frac{e^{ip \cdot x}}{(2\pi)^{3/2}} = \sqrt{\frac{2}{\pi}} \sum_{lm} i^l Y_{lm}^*(\hat{p}) Y_{lm}(\hat{x}) j_l(pr) \quad (3.2)$$

it follows that

$$\phi(x) = \sum_{lm} \int \frac{dE}{\sqrt{2E_p}} \psi^{Elm}(\mathbf{x}) e^{-iEt} a(Elm), \quad (3.3)$$

where

$$\psi^{Elm}(\mathbf{x}) = \sqrt{\frac{2pE}{\pi}} i^l Y_{lm}(\hat{x}) j_l(pr) \quad (3.4)$$

and $a(Elm)$ annihilates the state $|Elm\rangle$ [a spin $s=0$ case of Eq. (2.9)]. The spherical modes, ψ^{Elm} , are complete and may be superposed to make plane waves

$$\frac{e^{ip \cdot x}}{(2\pi)^{3/2}} = \frac{1}{\sqrt{pE}} \sum_{lm} Y_{lm}^*(\hat{p}) \psi^{Elm}(\mathbf{x}). \quad (3.5)$$

They are also orthonormal

$$\int d^3x \psi^{E'l'm'*}(\mathbf{x}) \psi^{Elm}(\mathbf{x}) = \delta_{l'l} \delta_{m'm} \delta(E' - E). \quad (3.6)$$

Now suppose that we have a collection of scalar fields, ϕ_i , interacting with a spherically symmetric, time independent background. We assume that the fields satisfy coupled, linear, Schrödinger-like equations. But, aside from rotational and time-translational symmetry, no other details of the dynamical equations need be specified. The fields may be expanded as

$$\phi_i(x) = \sum_j \sum_{lm} \int \frac{d^3p}{\sqrt{2E_p}} \psi_{ij}^{Elm}(\mathbf{x}) e^{-iEt} a_j(Elm), \quad (3.7a)$$

$$\psi_{ij}^{Elm}(\mathbf{x}) = \sqrt{\frac{2p_i E}{\pi}} i^l Y_{lm}(\hat{x}) \psi_l^{ij}(p_i, r). \quad (3.7b)$$

The assumed spherical symmetry of the background allows us to separate the original coupled partial differential equations for the fields ϕ_i . There results a set of coupled, radial, linear ordinary differential equations for the unknown functions $\psi_l^{ij}(p_i, r)$. Time independence of the background

ensures that these modes are stationary states. Note, however, that Eq. (3.7) holds even for fields ϕ_i of different masses m_i . In general, the modes will have different wave numbers, p_i , related by

$$\sqrt{p_i^2 + m_i^2} = E = \sqrt{p_j^2 + m_j^2}. \quad (3.8)$$

In general, there may be bound state modes—that is, square integrable modes with discrete energies $E < m_i$. The separation in Eqs. (3.7a) and (3.7b) still holds and can even describe unbound state-bound state transitions if there is an m_j with $m_j < E < m_i$. I will not explore this interesting possibility any further in this article. Instead, for simplicity, I will assume that there are no bound state modes. Further, I will assume that the background falls off fast enough as $r \rightarrow \infty$ so that the radial modes, $\psi_l^{ij}(p_i, r)$, are asymptotic to a sum of incoming and outgoing free radial modes

$$\lim_{r \rightarrow \infty} \psi_l^{ij}(p_i, r) = \frac{i}{2} [h_l^{(-)}(p_i r) \delta_{ij} - S_l^{ij}(E) h_l^{(+)}(p_i r)]. \quad (3.9)$$

We may always choose a basis such that the incoming modes are diagonal in the particle labeled i . The outgoing free modes then appear weighted by the scattering amplitudes $S_l^{ij}(E)$, as written in Eq. (3.9).

Define the modes

$$\psi_{ij}^{\text{in}}(\mathbf{p}, \mathbf{x}) = \frac{(2\pi)^{3/2}}{\sqrt{p_j E}} \sum_{lm} Y_{lm}^*(\hat{p}) \psi_{ij}^{E, lm}(\mathbf{x}), \quad (3.10)$$

which inherit orthonormality and completeness from the modes $\psi_{ij}^{E, lm}(\mathbf{x})$. These modes span the continuum states, labeled by the momentum $\mathbf{p}$. The fields ϕ_i may be expanded in terms of them as

$$\phi_i(x) = \sum_j \int \frac{d^3 p}{(2\pi)^{3/2} \sqrt{2E_p}} \psi^{ij}(\mathbf{p}, \mathbf{x}) e^{-iEt} a_j(\mathbf{p}). \quad (3.11)$$

In the context of a wave packet construction of the scattering process, one can prove that the following replacement is valid:¹³

$$Y_{lm}(\hat{x}) \frac{e^{ip'r - iEt}}{r} \rightarrow \frac{i}{2\pi E} \int d^3 p' \delta(E' - E) Y_{lm}(\hat{p}) e^{i\mathbf{p}' \cdot \mathbf{x} - iEt}. \quad (3.12)$$

Inserting Eq. (3.12) into Eq. (3.11) and using Eqs. (3.7b) and (3.9) yields

$$\lim_{t \rightarrow \infty} \phi_i(\mathbf{x}, t) = \sum_j \int \frac{d^3 p' d^3 p}{(2\pi)^{3/2} \sqrt{2E_p}} e^{i\mathbf{p}' \cdot \mathbf{x} - iEt} S_{\mathbf{p}'i, \mathbf{p}j} a_j(\mathbf{p}), \quad (3.13)$$

where

$$S_{\mathbf{p}'i, \mathbf{p}j} = \frac{1}{E \sqrt{p' p}} \delta(E' - E) \sum_{lm} Y_{lm}(\hat{p}') Y_{lm}^*(\hat{p}) S_l^{ij}(E). \quad (3.14)$$

Perhaps unsurprisingly, this expression for the two-body plane wave S matrix is exactly what one would expect from bra-ket reasoning and the transformation matrix

$$\langle \mathbf{p} s \sigma | E l m \rangle = \delta(E - E_p) \frac{1}{\sqrt{E_p}} Y_{lm}(\hat{p}). \quad (3.15)$$

More importantly, this S matrix transforms covariantly under the HLG, as the indices suggest, thereby ensuring that the field transforms as in Eq. (2.27).

We now do the same thing for the general (A, B) fields. First consider a free field

$$\phi_n(x) = \sum_{\sigma} \int \frac{d^3 p}{(2\pi)^{3/2} \sqrt{2E_p}} u_n(\mathbf{p}, s \sigma) e^{ip \cdot x} a(\mathbf{p}, s \sigma) \quad (3.16)$$

for a particle of spin s . Here the subscript n is shorthand for the compound index a, b and the wave function $u_n(\mathbf{p}, s \sigma)$ is given in Eq. (2.44). To resolve the $\hat{p}$ dependence of the factors in the integrand, use the expansion in Eq. (3.2) and insert the identity

$$1 = \int d^2 \hat{p}' \delta^2(\hat{p}' - \hat{p}) = \int d^2 \hat{p}' \sum_{lm} Y_{lm}^*(\hat{p}') Y_{lm}(\hat{p}) \quad (3.17)$$

into Eq. (3.16). Then interchange the $\hat{p}'$ and $\hat{p}$ integrations to get

$$\phi_n(x) = \sum_{JJ_z l} \int \frac{dE}{\sqrt{2E_p}} \psi_n^{EJJ_z ls}(\mathbf{x}) e^{-iEt} a(EJJ_z ls), \quad (3.18)$$

where

$$\psi_n^{EJJ_z ls}(\mathbf{x}) = \sqrt{\frac{2pE}{\pi}} \sum_{l'm'\sigma'} i^{l'} Y_{l'm'}(\hat{x}) j_{l'}(pr) \langle l's, J_z - \sigma', \sigma' | JJ_z \rangle u_n(E l' m' l, J_z - \sigma', s \sigma') \quad (3.19)$$

and

$$u_n(E l' m' l m s \sigma) = \int d^2 \hat{p} Y_{l'm'}^*(\hat{p}) Y_{lm}(\hat{p}) u_n(\mathbf{p}, s \sigma). \quad (3.20)$$

Before evaluating the integral in Eq. (3.20) explicitly, compare Eqs. (3.18)–(3.20) with Eqs. (3.3) and (3.4). The new feature that arises when $s > 0$ is that the wave functions, $u_n(E l' m' l m s \sigma)$, are not diagonal in orbital angular momentum. The mode, $\psi_n^{EJJ_z ls}(\mathbf{x})$, is made up of a *sum* of spherical harmonics, $Y_{l'm'}(\hat{x})$, weighted by $u_n(E l' m' l m s \sigma)$. Nevertheless, the state that corresponds to the mode, $\psi_n^{EJJ_z ls}(\mathbf{x})$, and that is annihilated by $a(EJJ_z ls)$ is sharp in orbital angular momentum l , as discussed in the previous section [see the discussion associated with Eqs. (2.8) and (2.9)]. Also note that, like the scalar field, the modes $\psi_n^{EJJ_z ls}(\mathbf{x})$ are orthonormal and complete.

Now to evaluate the integral in Eq. (3.20), first use the identity¹¹

$$Y_{lm}(\hat{p}) = D_{m0}^{(l)*}(\phi, \theta, 0) \quad (3.21)$$

and Eq. (2.44) to write $u_{ab}(E l' m' l m s \sigma)$ as a sum of integrals over $\hat{p}$ with integrands that are a product of five rotation matrices: three from the wave functions in Eq. (2.44) and two from the projection in Eq. (3.20). These may be combined by using the identity for combining rotation matrices

$$D_{m'n'}^{(J)*}(R)D_{mn}^{(J)}(R)=(-1)^{2(J-J')-m'-m-n'-n}\sqrt{(2J'+1)(2J+1)}\sum_K\begin{pmatrix} J' & J & K \\ m' & m & -m'-m \end{pmatrix} \\ \times\begin{pmatrix} J' & J & K \\ n' & n & -n'-n \end{pmatrix}D_{m'+m,n'+n}^{(K)}(R). \quad (3.22)$$

The fact that $\sigma'=a'+b'$ in Eq. (2.44) allows us to use the orthogonality result

$$\int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta D_{m'+a+b,a'+b'}^{(J')*}(\phi, \theta, 0) D_{m+\sigma,a'+b'}^{(J)}(\phi, \theta, 0) = \frac{4\pi}{2J+1} \delta_{J'J} \delta_{m'+a+b,m+\sigma} \quad (3.23)$$

to finally arrive at

$$u_{ab}(El'm'lms\sigma) = \delta_{m'+a+b,m+\sigma} \sum_{a'b'Js'} \sqrt{(2l'+1)(2l+1)(2s+1)(2s'+1)(2J+1)} e^{\theta(b'-a')} \\ \times (-1)^{A-B+m'+m+\sigma} \begin{pmatrix} A & B & s \\ a' & b' & -a'-b' \end{pmatrix} \begin{pmatrix} A & B & s' \\ a' & b' & -a'-b' \end{pmatrix} \\ \times \begin{pmatrix} A & B & s' \\ a & b & -a-b \end{pmatrix} \begin{pmatrix} l & s & J \\ m & \sigma & -m-\sigma \end{pmatrix} \begin{pmatrix} l & s & J \\ 0 & a'+b' & -a'-b' \end{pmatrix} \\ \times \begin{pmatrix} l' & s' & J \\ m' & a+b & -m'-a-b \end{pmatrix} \begin{pmatrix} l' & s' & J \\ 0 & a'+b' & -a'-b' \end{pmatrix}. \quad (3.24)$$

This explicit sum is finite and straightforward to evaluate, but is not particularly illuminating. In any case, the wave functions, $u_n(El'm'lms\sigma)$, rotate covariantly, as may be indirectly verified by using Eqs. (2.8) and (2.9) for the states and Eq. (2.31) for the field.

Finally, consider the case of a collection of higher spin, massive fields, ϕ_{n_i} , interacting with a spherically symmetric, time independent background. The fields may be expanded as

$$\phi_{n'}(x) = \sum_{\sigma\sigma'} \int \frac{d^3p}{(2\pi)^{3/2}\sqrt{2E_p}} \psi_{n'}^{(\text{in})}(\mathbf{p}\sigma\sigma';\mathbf{x}) e^{-iEt} a(\mathbf{p}\sigma\sigma'), \quad (3.25a)$$

$$\psi_{n'}^{(\text{in})}(\mathbf{p}\sigma\sigma',\mathbf{x}) = \frac{(2\pi)^{3/2}}{\sqrt{pE}} \sum_{lmJJ_z} Y_{lm}^*(\hat{p}) \langle lsm\sigma | JJ_z \rangle \sum_{l'} \psi_{n'}^{EJJ_zl's',ls}(\mathbf{x}), \quad (3.25b)$$

$$\psi_{n'}^{EJJ_zl's',ls}(\mathbf{x}) = \sum_{l''m''\sigma''} \sqrt{\frac{2p'E}{\pi}} i^{l''} Y_{l''m''}(\hat{x}) \psi_{l''}^{EJJ_zl's',ls}(p',r) \\ \times \langle l's',J'_z-\sigma'',\sigma'' | JJ_z \rangle u_{n'}(El''m''l',J'_z-\sigma'',s'\sigma''), \quad (3.25c)$$

$$\lim_{r \rightarrow \infty} \psi_{l''}^{EJJ_zl's',ls}(p',r) = \frac{i}{2} [h_{l''}^{(-)}(p'r) \delta_{l',l} \delta_{s',s} - S_{l',s',ls}^J(E) h_{l''}^{(+)}(p'r)]. \quad (3.25d)$$

The expansion given in Eqs. (3.25a)–(3.25d) is a straightforward generalization of Eqs. (3.11), (3.10), (3.7b), and (3.9), respectively. Again, for simplicity, the background is assumed not to admit any bound state modes. That is, it is assumed to fall off fast enough as $r \rightarrow \infty$ so that the

modes are asymptotic to a sum of free modes. The notation in Eq. (3.25) is cumbersome and even then, some things are implicit or warrant further comment. The primed variables refer to the outgoing quantum numbers and the unprimed variables to the incoming quantum numbers [in the sense of a wave packet construction as in Eqs. (3.12)–(3.14) and (3.26)–(3.28) below]. I have not included the additional i, j indices that were used to label the fields in the scalar case; this is left implicit in Eq. (3.25). Again, inelasticity is accounted for in the sense of Eq. (3.8). The separation of variables is written for each orthonormal mode, $\psi_{n'}^{EJJ_z l' s', ls}(\mathbf{x})$, in Eq. (3.25c). Note that this mode accounts for transitions $l, s \rightarrow l', s'$. We may reassemble these modes as in Eq. (3.25b) to get the asymptotic plane wave modes, $\psi_n^{(in)}(\mathbf{p} s \sigma; \mathbf{x})$. The coupled partial differential equations for the ϕ_n give rise to a matrix of ordinary differential equations for the coupled radial modes, $\psi_{l''}^{EJJ_z l' s', ls}(p', r)$. We may choose the solutions so that the incoming waves, $h^{(-)}(p' r) \delta_{l' l} \delta_{s' s}$, are diagonal in l, s (and implicitly in field type i). We then automatically get the unitary partial wave S -matrix amplitudes, $S_{l' s', ls}^J(E)$, as the coefficients of the outgoing waves in Eq. (3.25d). In particular, note that the amplitudes, $S_{l' s', ls}^J(E)$, must be independent of l'' in order that the radial mode, $\psi_{l''}^{EJJ_z l' s', ls}(p', r)$, reduce to a sum of free modes. Equations (3.25a)–(3.25d) constitute the answer to the question posed in the introduction on how the partial wave S matrix is encoded in the higher spin fields—at least for $m > 0$.

The superscript, in, indicates that the superposition of spherical modes in Eq. (3.25b), for $\psi^{(in)}(\mathbf{p} s \sigma; \mathbf{x})$, is such that at very early times $\psi^{(in)}(\mathbf{p} s \sigma; \mathbf{x})$ describes an incoming pure plane wave with momentum $\mathbf{p}$, spin s , and spin $\hat{z}$ component σ . At late times, $\psi^{(in)}(\mathbf{p} s \sigma; \mathbf{x})$ describes a collection of outgoing waves with momentum $\mathbf{p}$, spin s' and $\hat{z}$ component σ' . To make this more precise, use the wave packet result¹³

$$\begin{aligned} \sum_{l'' m''} Y_{l'' m''}(\hat{x}) \frac{e^{i p' r - i E t}}{r} u_{n'}(E l'' m'' l' m' s' \sigma') \\ \rightarrow \frac{i}{2 \pi E} \int d^3 p' \delta(E' - E) Y_{l' m'}(\hat{p}') e^{i \mathbf{p}' \cdot \mathbf{x} - i E t} u_{n'}(\mathbf{p}' s' \sigma'). \end{aligned} \quad (3.26)$$

Putting Eq. (3.26) into Eq. (3.25) we get

$$\lim_{t \rightarrow \infty} \phi_{n'}^{s'}(\mathbf{x}, t) = \sum_{s \sigma \sigma'} \int \frac{d^3 p' d^3 p}{(2 \pi)^{3/2} \sqrt{2 E_p}} u_{n'}(\mathbf{p}' s' \sigma') e^{i \mathbf{p}' \cdot \mathbf{x} - i E t} S_{\mathbf{p}' s' \sigma', \mathbf{p} s \sigma} a(\mathbf{p} s \sigma), \quad (3.27)$$

where

$$S_{\mathbf{p}' s' \sigma', \mathbf{p} s \sigma} = \frac{1}{E \sqrt{p' p}} \sum_{JJ_z l' m' l m} Y_{l' m'}(\hat{p}') \langle l' m' s' \sigma' | JJ_z \rangle S_{l' s', ls}^J(E) \delta(E' - E) \langle JJ_z | l m s \sigma \rangle Y_{lm}^*(\hat{p}). \quad (3.28)$$

As in the scalar field case, Eq. (3.14), the two-body S matrix in Eq. (3.28) is what we expect from bra–ket reasoning and the result

$$\langle \mathbf{p} s \sigma | EJJ_z l s \rangle = \delta(E - E_p) \frac{1}{\sqrt{E_p}} \sum_m \langle l s m \sigma | JJ_z \rangle Y_{lm}(\hat{p}). \quad (3.29)$$

This S matrix transforms covariantly, thereby ensuring that the fields ϕ_n transform properly. Any given problem with relativistic fields is reduced, by the separation of variables in Eq. (3.25), to a

computation of coupled radial ordinary differential equations for the modes $\psi_{l'n'}^{EJJ_zl's',is}(p',r)$. Rotational symmetry has allowed the problem to be separated so that the dynamics is contained entirely in $S_{l's',is}^J(E)$.

IV. SPHERICAL WAVE EXPANSION FOR $m=0$ FIELDS

As in the massive case, the plane wave expansion for free massless fields may be used to infer the corresponding SWE. The arguments for the $m=0$ expansion and its extension to background scattering proceed mostly as in the last section. The only detailed differences occur in the transformation between plane wave and angular momentum sharp states, and in the wave functions. That is, Eq. (2.9) must be replaced by Eq. (2.27), and Eq. (2.44) by Eq. (2.50).

The subsequent analysis results in a massless extension of Eqs. (3.25a)–(3.25d). That is, imagine that we have a collection of massless fields interacting with a spherically symmetric background. And imagine that there are no bound state waves in this background. Then the fields may be expanded as

$$\phi_{n'}(x) = \sum_{\lambda} \int \frac{d^3p}{(2\pi)^{3/2} \sqrt{2p}} \psi_{n'}^{(in)}(\mathbf{p}\lambda; \mathbf{x}) e^{-iEt} a(\mathbf{p}\lambda), \quad (4.1a)$$

$$\psi_{n'}^{(in)}(\mathbf{p}\lambda, \mathbf{x}) = \frac{(2\pi)^{3/2}}{p} \sum_{JJ_z} \sqrt{\frac{2J+1}{4\pi}} D_{J_z\lambda}^{(J)}(R(\hat{p})) \psi_{n'}^{JJ_z\lambda',\lambda}(\mathbf{x}), \quad (4.1b)$$

$$\psi_{n'}^{JJ_z\lambda',\lambda}(\mathbf{x}) = \sum_{lm} \sqrt{\frac{2}{\pi}} p i^l Y_{lm}(\hat{x}) \psi_l^{JJ_z\lambda',\lambda}(p, r) u_{n'}(p l m J J_z, \lambda'), \quad (4.1c)$$

$$\lim_{r \rightarrow \infty} \psi_l^{JJ_z\lambda',\lambda}(p, r) = \frac{i}{2} [h_l^{(-)}(pr) \delta_{l',l} \delta_{s's} - S_{\lambda',\lambda}^J(p) h_l^{(+)}(pr)], \quad (4.1d)$$

where n, n' are compound indices $[(a, b)$ and (a', b') , respectively] and

$$u_{ab}(p l m J J_z \lambda) = \frac{\sqrt{(2J+1)(2J'+1)}}{4\pi} \int d^2\hat{p} D_{J_z\lambda}^{(J)*}(R(\hat{p})) D_{m0}^{(I)}(R(\hat{p})) u_{ab}(\mathbf{p}, \lambda). \quad (4.2)$$

Equation (4.1a) involves an expansion in terms of in modes which, in a wave packet sense, describe incoming pure plane waves with momentum $\mathbf{p}$ and helicity λ . Equation (4.1b) expands the plane wave in modes, $\psi_{n'}^{(in)}(\mathbf{p}\lambda, \mathbf{x})$ in terms of angular momentum-sharp in modes, $\psi_{n'}^{JJ_z\lambda',\lambda}(\mathbf{x})$. These modes may be factored as in Eq. (4.1c). For modes propagating in spherically symmetric backgrounds, this factorization separates the original, Schrödinger-like equations. The resulting radial modes $\psi_l^{JJ_z\lambda',\lambda}(p, r)$, satisfy coupled, linear ordinary differential equations. The boundary conditions on these modes are first that they be regular at the origin and, second, that they involve helicity-diagonal incoming waves. The outgoing waves will then generally be non-diagonal in helicity and involve the scattering amplitudes, $S_{\lambda',\lambda}^J(p)$, as in Eq. (4.1d). Note that these amplitudes must be independent of l for the modes to be free as $r \rightarrow \infty$.

The in mode, $\psi_{n'}^{(in)}(\mathbf{p}\lambda, \mathbf{x})$, describes an incoming particle with momentum $\mathbf{p}$ and helicity λ which then subsequently scatters off the background into one of a variety of outgoing particles labeled by momentum $\mathbf{p}$ and helicity λ' . As in the massive case in Sec. III, this interpretation holds in the context of a wave packet construction. Equation (3.26) is replaced by

$$\sum_{lm} Y_{lm}(\hat{x}) \frac{e^{ipr - iEt}}{r} u_{n'}(plmJJ_z, \lambda') \rightarrow \frac{i}{2\pi p} \int d^3p' \delta(p' - p) \sqrt{\frac{2J+1}{4\pi}} D_{J_z\lambda'}^{(J)*}(R(\hat{p}')) \times e^{i\mathbf{p}' \cdot \mathbf{x} - ip't} u_{n'}(\mathbf{p}'\lambda'). \quad (4.3)$$

Then Eqs. (4.1a)–(4.1d) and (4.3) combine to give

$$\lim_{t \rightarrow \infty} \psi_{n'}^{\text{in}}(\mathbf{p}\lambda, \mathbf{x}) e^{-ip't} = \sum_{\lambda'} \int d^3p' u_{n'}(\mathbf{p}', \lambda') e^{i\mathbf{p}' \cdot \mathbf{x} - ip't} S_{\mathbf{p}'\lambda', \mathbf{p}\lambda}, \quad (4.4)$$

where

$$S_{\mathbf{p}'\lambda', \mathbf{p}\lambda} = \sum_{JJ_z} \frac{1}{p} \sqrt{\frac{2J+1}{4\pi}} D_{J_z\lambda}^{(J)}(R(\hat{p}')) S_{\lambda', \lambda}^J(p) \delta(p' - p) \frac{1}{p} \sqrt{\frac{2J+1}{4\pi}} D_{J_z\lambda}^{(J)}(R(\hat{p})). \quad (4.5)$$

This S matrix is what we would expect from bra–ket reasoning using the transformation between momentum-sharp and angular momentum-sharp states in Eq. (2.27).

The results in Eqs. (3.25a)–(3.25d) and (4.1a)–(4.1d) may be combined to account for transitions amongst a collection of $m \geq 0$ particles scattering off a background.

Finally, the wave functions defined in Eq. (4.2) may be written in terms of a product of $3j$ symbols. With the result in Eq. (2.50), the integral over $\hat{p}$ in Eq. (4.2) may be evaluated to yield

$$u_{ab}(plmJJ_z\lambda) = \left(\frac{p}{\kappa}\right)^{A+B} \sqrt{(2J+1)(2l+1)} (-)^{A+a} \sum_K (2K+1) \begin{pmatrix} J & A & K \\ J_z & -a & a+J_z \end{pmatrix} \times \begin{pmatrix} J & A & K \\ \lambda & A & -B \end{pmatrix} \begin{pmatrix} l & B & K \\ m & b & a-J_z \end{pmatrix} \begin{pmatrix} l & B & K \\ 0 & B & -\lambda-A \end{pmatrix}. \quad (4.6)$$

As in the $m > 0$ case, Eq. (3.24), the result is not in itself illuminating. The salient facts are first, for any (A, B) field for a helicity λ particle the wave functions, $u_{ab}(plmJJ_z\lambda)$, may be computed by the finite sum in Eq. (4.6); and second, these wave functions rotate covariantly as may be indirectly verified using Eq. (2.25) and the $m=0$ realization of Eq. (2.31).

V. SPHERICAL WAVE EXPANSION FOR $m=0$ POTENTIALS

The results (3.25a)–(3.25d) and (4.1a)–(4.1d) hold generally for $m \geq 0$ particles whose interactions are described by Lorentz covariant fields. The neutrino, for example, is accounted for by a massless $(A, B) = (0, 1/2)$ field, and its SWE may be deduced from Eqs. (4.1a)–(4.1d). However, the other known massless particles—the photon and the graviton—have interactions that are not described by Lorentz covariant fields. This can be guessed from the factor in Eq. (2.50)

$$\left(\frac{E}{\kappa}\right)^{A+B} = \left(\frac{E}{\kappa}\right)^{\lambda+2 \min(A, B)}. \quad (5.1)$$

If photons and gravitons were described by interacting fields alone, then their interactions would vanish in the long wavelength limit. For example, the photon has helicity $\lambda=1$, and from Eq. (5.1) there would therefore be no Thompson scattering. Further, Eq. (5.1) implies that the potential due to massless, helicity $\pm\lambda$, particle exchange would go as $(1/r)^{1+\lambda+4 \min(A, B)}$, and there would be no long range, $1/r$ Coulomb potential.

Instead, nature chooses an alternative where the interactions are mediated by gauge potentials. These potentials do *not* transform covariantly. Recall that this is the basic requirement (2.31) for fields. Instead, a Lorentz transformation is realized by a covariant rotation of the potential plus an extra “gauge transformation” contribution.¹⁴ Weinberg⁸ describes this alternative in detail. I will just state the main results relevant to a SWE. In way of motivation, first observe that an integer, helicity $\pm\lambda$, particle admits an alternative, vector representation for its $(A, B) = (0, \lambda)$ or $(\lambda, 0)$ field⁸

$$F_{\pm}^{[\mu_1 \nu_1] \cdots [\mu_\lambda \nu_\lambda]}(x) = i^\lambda \int \frac{d^3 p}{(2\pi)^{3/2} \sqrt{2p}} [p^{\mu_1} e_{\pm}^{\nu_1}(\mathbf{p}) - p^{\nu_1} e_{\pm}^{\mu_1}(\mathbf{p})] \cdots [p^{\mu_\lambda} e_{\pm}^{\nu_\lambda}(\mathbf{p}) - p^{\nu_\lambda} e_{\pm}^{\mu_\lambda}(\mathbf{p})] \\ \times [a(\mathbf{p}, \pm\lambda) e^{i\mathbf{p} \cdot \mathbf{x}} + (-)^\lambda a^*(\mathbf{p}, \mp\lambda) e^{-i\mathbf{p} \cdot \mathbf{x}}]. \quad (5.2)$$

Here $a(\mathbf{p}, \pm\lambda)$ destroys the massless particle with momentum $\mathbf{p}$ and helicity $\pm\lambda$. For simplicity, I have assumed that the particle is its own antiparticle (as is true for the photon and graviton), and defined their relative phases to simplify the potential below in Eq. (5.8). The polarization vectors in the expansion in Eq. (5.2) are defined by

$$e_{\pm}^{\mu}(\mathbf{p}) = \mathcal{Z}_{\nu}^{\mu}(\mathbf{p}/\kappa) e_{\pm}^{\nu}, \quad (5.3)$$

where

$$e_{\pm}^1 = \mp \frac{1}{\sqrt{2}}, \quad e_{\pm}^2 = -\frac{i}{\sqrt{2}}, \quad e_{\pm}^3 = 0 = e_{\pm}^0 \quad (5.4)$$

and $\mathcal{Z}_{\nu}^{\mu}(\mathbf{p}/\kappa)$ is the four-vector representation of $\mathcal{Z}(\mathbf{p}/\kappa)$ defined in Eqs. (2.22)–(2.24).

With these definitions, the field in Eq. (5.2) may be shown directly to transform covariantly as in Eq. (2.31). Alternatively, following Secs. III and IV, the wave function

$$u_{\pm}^{[\mu_1 \nu_1] \cdots [\mu_\lambda \nu_\lambda]}(\mathbf{p}, \lambda) \equiv [p^{\mu_1} e_{\pm}^{\nu_1}(\mathbf{p}) - p^{\nu_1} e_{\pm}^{\mu_1}(\mathbf{p})] \cdots [p^{\mu_\lambda} e_{\pm}^{\nu_\lambda}(\mathbf{p}) - p^{\nu_\lambda} e_{\pm}^{\mu_\lambda}(\mathbf{p})] \quad (5.5)$$

may be derived by solving the covariance condition, Eq. (2.45), for the case of an irreducible 2λ -indexed tensor representation of the HLG.

The polarization vectors in Eq. (5.3) only depend on $\hat{p}$ —the direction of $\mathbf{p}$. Indeed, it follows from Eq. (5.4) and from the definition of $\mathcal{Z}(\mathbf{p}/\kappa)$ in Eqs. (2.22)–(2.24) that

$$e_{\pm}^{\mu}(\mathbf{p}) = R_{\nu}^{\mu}(\mathbf{p}/\kappa) e_{\pm}^{\nu} = e_{\pm}^{\mu}(\hat{p}). \quad (5.6)$$

In fact, Eqs. (5.4) and (5.6) imply that

$$e_{\pm}^0(\hat{p}) = 0, \quad \hat{p} \cdot \mathbf{e}_{\pm}(\hat{p}) = 0 \quad (5.7)$$

and the $e_{\pm}^{\mu}(\mathbf{p})$ are thus the usual transverse, circular polarization vectors.

Now *define* the potential $A_{\pm}^{\mu_1 \cdots \mu_\lambda}(x)$ corresponding to the field $F_{\pm}^{[\mu_1 \nu_1] \cdots [\mu_\lambda \nu_\lambda]}(x)$ by

$$A_{\pm}^{\mu_1 \cdots \mu_\lambda}(x) \equiv \int \frac{d^3 p}{(2\pi)^{3/2} \sqrt{2p}} e_{\pm}^{\nu_1}(\mathbf{p}) \cdots e_{\pm}^{\mu_\lambda}(\mathbf{p}) [a(\mathbf{p}, \pm\lambda) e^{i\mathbf{p} \cdot \mathbf{x}} + a^*(\mathbf{p}, \mp\lambda) e^{-i\mathbf{p} \cdot \mathbf{x}}]. \quad (5.8)$$

The obvious relation of the potentials and the fields is that $F_{\pm}^{[\mu_1 \nu_1] \cdots [\mu_\lambda \nu_\lambda]}$ are some λ th derivatives of $A_{\pm}^{\mu_1 \cdots \mu_\lambda}$. For example,

$$F_{\pm}^{[\mu \nu]}(x) = \partial^{\mu} A^{\nu}(x) - \partial^{\nu} A^{\mu}(x),$$

$$F_{\pm}^{[\mu\nu][\lambda\eta]}(x) = \partial^{\mu}\partial^{\lambda}A^{\nu\eta}(x) - \partial^{\mu}\partial^{\eta}A^{\nu\lambda}(x) - \partial^{\nu}\partial^{\lambda}A^{\mu\eta}(x) + \partial^{\nu}\partial^{\eta}A^{\mu\lambda}(x). \quad (5.9)$$

The potentials, $A_{\pm}^{\mu_1\cdots\mu_{\lambda}}$, are not covariant fields. To see how they transform, first note that Eq. (5.3) implies that

$$\Lambda_{\nu}^{\mu}e_{\pm}^{\nu}(\mathbf{p}) = e^{\pm i\Theta[\mathcal{L}^{-1}(\Lambda\mathbf{p})\Lambda\mathcal{Z}(\mathbf{p})]}e_{\pm}^{\mu}(\Lambda\mathbf{p}) + X_{\pm}[\mathcal{L}^{-1}(\Lambda\mathbf{p})\Lambda\mathcal{Z}(\mathbf{p})](\Lambda p)^{\mu}. \quad (5.10)$$

Here Θ is defined as in Eq. (2.25), and the $X_{\pm}$ are generated by $L_1 \pm iL_2$ [defined in Eq. (2.18)] in the Lorentz four-vector representation. But then Eq. (5.10) may be used to prove that

$$U(\Lambda)A_{\pm}^{\mu_1\cdots\mu_{\lambda}}(x)U^{-1}(\Lambda) = \Lambda_{\nu_1}^{\mu_1}\cdots\Lambda_{\nu_{\lambda}}^{\mu_{\lambda}}A_{\pm}^{\nu_1\cdots\nu_{\lambda}}(\Lambda x) + \sum_{r=1}^{\lambda}\partial^{\mu_r}\Phi_{\pm}^{\mu_1\cdots\mu_{r-1}\mu_{r+1}\cdots\mu_{\lambda}}(x;\Lambda), \quad (5.11)$$

where $\Phi_{\pm}$ are well-defined functions of x and Λ . Compare Eq. (5.11) with the covariance condition (2.31) on fields. A Lorentz transformation on the particle states is realized as a combined Lorentz transformation and “gauge transformation”¹⁴ on the potential. Note that Eq. (5.11) implies that $F_{\pm}$ are Lorentz tensors since the gauge terms cancel upon antisymmetrization.

The plane wave expansion of the free potentials in Eq. (5.8) immediately gives the following SWE:

$$A_{\pm}^{i_1\cdots i_{\lambda}}(x) = \sum_{JJ_z} \int \frac{dp}{\sqrt{2p}} \psi^{i_1\cdots i_{\lambda}}(pJJ_z, \pm\lambda; \mathbf{x}) e^{-ipx} a(pJJ_z, \pm\lambda), \quad (5.12)$$

where

$$\psi^{i_1\cdots i_{\lambda}}(pJJ_z, \pm\lambda; \mathbf{x}) = \sum_{lm} \sqrt{\frac{2}{\pi}} p i^l Y_{lm}(\hat{x}) j_l(pr) u_{\pm}^{i_1\cdots i_{\lambda}}(plmJJ_z, \pm\lambda) \quad (5.13)$$

and the wave functions are given by

$$\begin{aligned} u_{\pm}^{i_1\cdots i_{\lambda}}(plmJJ_z, \pm\lambda) &\equiv \int d^3p \sqrt{\frac{2J+1}{4\pi}} D_{J_z, \pm\lambda}^{(J)*}(R(\hat{p})) Y_{lm}^*(\hat{p}) e_{\pm}^{i_1}(\hat{p}) \cdots e_{\pm}^{i_{\lambda}}(\hat{p}) \\ &= \sqrt{\frac{2l+1}{2J+1}} \langle 1, 1, -i_1, -i_2 | 2, -(i_1+i_2) \rangle \\ &\quad \times \cdots \times \left\langle \lambda-1, 1, -\sum_{j=1}^{\lambda-1} i_j, i_{\lambda} \middle| \lambda, -\sum_{j=1}^{\lambda} i_j \right\rangle \\ &\quad \times \left\langle l\lambda m, -\sum_{j=1}^{\lambda} i_j \middle| JJ_z \right\rangle \langle l\lambda 0, \pm\lambda | J, \pm\lambda \rangle (-)^{\lambda}. \end{aligned} \quad (5.14)$$

The integral in Eq. (5.14) has been easily evaluated by using the explicit result

$$e_{\pm}^i(\hat{p}) = D_{-i, \pm 1}^{(1)}(R(\hat{p})). \quad (5.15)$$

Equations (5.12)–(5.15) and the vanishing of the gauge potential whenever any of its Lorentz indices equals zero then completely specifies the SWE for an arbitrary helicity-free potential.

Now suppose that the massless particles interact, via gauge potentials, with a spherically symmetric background. Weinberg proved in Ref. 8 that the underlying field theory is gauge invariant if the potentials are coupled only to conserved currents. We will thus assume the theory to be gauge invariant and imagine that the calculations are done in the Coulomb or radiation gauge. For example, for the $\lambda=1$ photon, we assume

$$\nabla \cdot \mathbf{A} = 0. \quad (5.16)$$

With this choice, the time component of the gauge potential may be written in terms of the instantaneous charge density⁸

$$A^0(\mathbf{x}, t) = \int \frac{J^0(\mathbf{y}, t)}{4\pi|\mathbf{x}-\mathbf{y}|} d^3y. \quad (5.17)$$

Thus, the only components of the potential that are needed to encode scattering data are the transverse, spatial components. This result generalizes to arbitrary λ , and so the free potential SWE in Eqs. (5.12)–(5.14) immediately generalizes to include background scattering when the theory is formulated in the Coulomb gauge. The spherical wave expansion in this case may be written as in Eqs. (4.1a)–(4.1d) with the replacements

$$\phi_{n'} \rightarrow A_{\pm}^{i_1 \cdots i_\lambda}, \quad u_{n'}(plmJJ_z, \lambda') \rightarrow u_{\pm}^{i_1 \cdots i_\lambda}(plmJJ_z, \pm \lambda'). \quad (5.18)$$

VI. DISCUSSION AND CONCLUSIONS

Although the reasoning in this article is straightforward enough, the results might seem unfamiliar and recondite. It is thus useful to touch base with two well-known SWEs in specific cases—the photon, and the Dirac spin-1/2 electron. For the photon, the expansion in Eqs. (5.12)–(5.14) may be directly shown to reduce to the usual vector spherical harmonic expansion of the electromagnetic gauge potential. When $\lambda=1$, Eqs. (5.12)–(5.14) imply

$$A_{\pm}^i(\mathbf{x}) = \sum_{JM} \int \frac{dp}{\sqrt{2p}} a(pJM, \pm 1) e^{-ipt} (-) \sum_{lm} \sqrt{\frac{2(2l+1)}{\pi(2J+1)}} i^l p Y_{lm}(\hat{x}) j_l(pr) \langle l1m, -i|JM \rangle \\ \times \langle l10, \pm 1|J, \pm 1 \rangle + \text{h.c.}, \quad (6.1)$$

where h.c. stands for the negative frequency, Hermitian conjugate part of the potential. But then Eq. (6.1) and the identity [see Eq. (7.6.33) in Ref. 11]

$$e^{i\mathbf{p} \cdot \mathbf{x}} \mathbf{e}_i \cdot \mathbf{e}_{\pm}(\hat{p}) = \sum_{lm} i^{l-1} \sqrt{2\pi(2l+1)} D_{m\lambda}^{(l)}(R(\hat{p})) \mathbf{e}_i \cdot \left[-\frac{i}{p} \nabla \times j_l(pr) \mathbf{Y}_{lm}(\mathbf{x}) \mp i j_l(pr) \mathbf{Y}_{lm}(\mathbf{x}) \right], \quad (6.2)$$

where $\mathbf{Y}_{LM}(\mathbf{x})$ are the usual vector spherical harmonics,¹¹ may be combined to show that Eq. (6.1) is equivalent to

$$\mathbf{A}_{\pm}(\mathbf{x}) = \sum_{JM} \int \frac{dp}{\sqrt{2p}} a(pJM, \pm 1) e^{-ipt} \frac{i^{J-1} p}{\sqrt{\pi}} \left[\mp i j_J(pr) \mathbf{Y}_{JJM}(\mathbf{x}) + \frac{1}{ip} \nabla \times j_J(pr) \mathbf{Y}_{JJM}(\mathbf{x}) \right] \\ + \text{h.c.} \quad (6.3)$$

Equation (6.3) is the usual vector spherical harmonic expansion of the gauge potential in the Coulomb gauge.¹⁵

Next consider the spin-1/2 Dirac electron, which is described by a reducible, $(1/2,0)+(0,1/2)$ field. The irreducible $(1/2,0)$ and $(0,1/2)$ fields are left and right handed Weyl fields, respectively. They may be separately expanded in spherical waves by applying the results of Sec. III, and then the expansions may be directly summed to give the SWE for the Dirac field. In particular, the free field expansions are given by Eqs. (3.24) and (3.25a)–(3.25d) by taking $S_{l',\sigma',l\sigma}^J = 1$. A lengthy calculation shows that the spherical waves for the $(1/2,0)$ field are given by

$$\begin{aligned} \psi_a^{EJJ_z, l=J\pm 1/2, s=1/2}(\mathbf{x}) = & \sqrt{\frac{pE(E+m)}{2\pi m}} i^{J\pm 1/2} \left(j_{J\pm 1/2}(pr) (\chi_{J, J\pm 1/2}^{J_z}(\hat{x}))_a \right. \\ & \left. \mp \frac{p}{E+M} j_{J\mp 1/2}(pr) (\chi_{J, J\mp 1/2}^{J_z}(\hat{x}))_a \right). \end{aligned} \quad (6.4)$$

Here, $\chi_{J,l}^{J_z}(\hat{x})$ are spinor spherical harmonics

$$(\chi_{J,l}^{J_z}(\hat{x}))_a \equiv \sum_{m,\sigma} \langle l, 1/2, m\sigma | JJ_z \rangle Y_{lm}(\hat{x}) (\chi_m)_a, \quad (6.5)$$

where $\chi_{\pm}$ are the usual spinors

$$(\chi_m)_a = \delta_{ma}, \quad m, a = \pm \frac{1}{2}. \quad (6.6)$$

Simply changing the sign in the second term of Eq. (6.4) gives the spherical waves for the free $(0,1/2)$ field, ψ_b . Further, it may be verified that the operator $\boldsymbol{\sigma} \cdot \mathbf{L} + 1$ acting on each of these waves gives the other times the constant $-\kappa \equiv \pm(J+1/2)$

$$(\boldsymbol{\sigma} \cdot \mathbf{L} + 1) \psi_{(0,1/2)} = -\kappa \psi_{(1/2,0)}, \quad (\boldsymbol{\sigma} \cdot \mathbf{L} + 1) \psi_{(1/2,0)} = -\kappa \psi_{(0,1/2)}. \quad (6.7)$$

Now the Dirac field, ψ_{ab} , is composed of the two Weyl fields by a direct sum which may be written as

$$\psi_{ab} = \begin{pmatrix} \psi_a \\ \psi_b \end{pmatrix}. \quad (6.8)$$

Equations (6.4)–(6.8) are equivalent to the expansion of the Dirac field in Ref. 3. Furthermore, in the basis¹⁶ chosen in Eq. (6.8) the Dirac matrix β is given by

$$\beta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (6.9)$$

But then Eqs. (6.7) and (6.9) combine to imply that the operator $K = \beta(\boldsymbol{\sigma} \cdot \mathbf{L} + 1)$ is diagonal in the spherical basis¹⁶

$$K \begin{pmatrix} \psi_a \\ \psi_b \end{pmatrix} = -\kappa \begin{pmatrix} \psi_a \\ \psi_b \end{pmatrix}. \quad (6.10)$$

K remains diagonal for the modes in a spherically symmetric background. This is now seen to be a consequence of the reducibility of the Dirac field and Eq. (6.7).

The conventional arguments for the SWE of the gauge potential and the Dirac electron field do not easily generalize to higher spin fields. We now see that the construction in this article, which holds for any spin or helicity, recovers these two well-known expansions.

The same strategy is used to identify the SWEs in each of the Secs. III–V. First Poincaré covariance is used to construct plane wave expansions of free fields for particles of arbitrary spin

and mass, $m \geq 0$. (Similarly, Poincaré and gauge covariances are used for the construction of gauge potentials for massless particles of arbitrary helicity.) Second, the Rayleigh SWE of plane waves, Eq. (3.2), is combined with the transformation between momentum and angular momentum-sharp states, Eqs. (2.9) and (2.27), to infer the free field SWE's in Eqs. (3.3), (3.18), (4.1), and (5.12). Furthermore, as emphasized in the text, these expansions involve a separation of variables that is valid whenever the fields or potentials interact with a spherically symmetric background. In each of the Secs. III–V, I worked out the SWE for backgrounds that support no bound modes. Only scattering was assumed to occur. The expansions in Eqs. (3.25), (4.1), and (5.18) give explicit prescriptions for computing the S matrix for particles scattering off the background. The computation is reduced to integrating radial modes out to asymptotic distances and identifying the coefficients of the outgoing modes [see Eqs. (3.25d) and (4.1d)]. This amounts to a relativistic generalization of the spherical potential scattering problem in nonrelativistic quantum mechanics.

There are several possible extensions and modifications of this analysis. For example, the wave functions of the mass $m > 0$ particles may also be projected onto the helicity basis of Jacob and Wick¹⁷ by following the reasoning in this article applied to that representation. For another example, both in and out modes are needed to construct the full relativistic field theory [see Eq. (2.51)]. This can be heuristically understood by recalling that antiparticles may be thought of as particles propagating backward in time. Their corresponding scattering modes thus involve a complicated incoming spherical wave that scatters into an outgoing pure plane wave. A more precise motivation for computing both in and out modes is that the Feynman boundary condition Green functions involve products of both sets of modes. The relation of the out and in modes is the same as the relation of the advanced and retarded modes in nonrelativistic quantum mechanics. For time reversal invariant theories, they involve the same scattering data. In any case, the in and out modes are separately complete.

Another possible extension would be to apply the basic strategy in this article to identify cylindrical wave expansions for problems with axial symmetry. Again the starting point is the plane wave expansion of the free fields and potentials. The two necessary new ingredients would be the cylindrical wave expansion of plane waves and the transformation between momentum states and cylindrical states ($|Ep_z M, s\rangle$ where M is the eigenvalue of J_z and s is the spin).

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¹See any textbook on quantum mechanics. For example, see E. Merzbacher, *Quantum Mechanics* (Wiley, New York, 1970).

²See, for example, Sec. 3–8 in J. J. Sakurai, *Advanced Quantum Mechanics* (Addison-Wesley, Reading, MA, 1980).

³See, for example, Sec. 33 in M. E. Rose, *Relativistic Electron Theory* (Wiley, New York, 1961).

⁴R. Rajaraman, *Solitons and Instantons* (North-Holland, New York, 1987).

⁵A three dimensional example involving pion–nucleon scattering in the Skyrme model is worked out in detail in M. P. Mattis and M. Karliner, *Phys. Rev. D* **31**, 2833 (1985).

⁶For other examples involving relativistic scattering off external potentials, see M. L. Goldberger and K. M. Watson, *Collision Theory* (Wiley, New York, 1964).

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¹¹We follow the definitions and conventions in L. C. Beidenharn and J. D. Louck, in *Angular Momentum in Quantum Physics* (Addison-Wesley, Reading, MA, 1981).

- ¹²The spherical Bessel functions including $j_l(z)$, are defined following the conventions in Ref. 7.
- ¹³Nice discussions which include proofs of equivalent identities occur in Chap. 11 of Ref. 1 and in Chap. 10 of Ref. 7.
- ¹⁴See any textbook on gauge theories. For example, see Chaps. 3 and 4 in C. Quigg, *Gauge Theories of the Strong, Weak, and Electromagnetic Interactions* (Addison-Wesley, Reading, MA, 1983).
- ¹⁵In addition to Sec. 6 of Chap. 7 in Ref. 11, see Chap. 16 in J. D. Jackson, *Classical Electrodynamics* (Wiley, New York, 1980).
- ¹⁶Dirac particles in spherical backgrounds and, in particular, the operator $K = \beta(\boldsymbol{\sigma} \cdot \mathbf{L} + 1)$, are discussed in Sec. 3.7 in Ref. 2 and in Chap. 5 of Ref. 3.
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